

DepthColorLinked_005m — Design & Stimulus

Factor 1 — Dot Cueing (✓/✗): does the delayed-onset dot field translate?

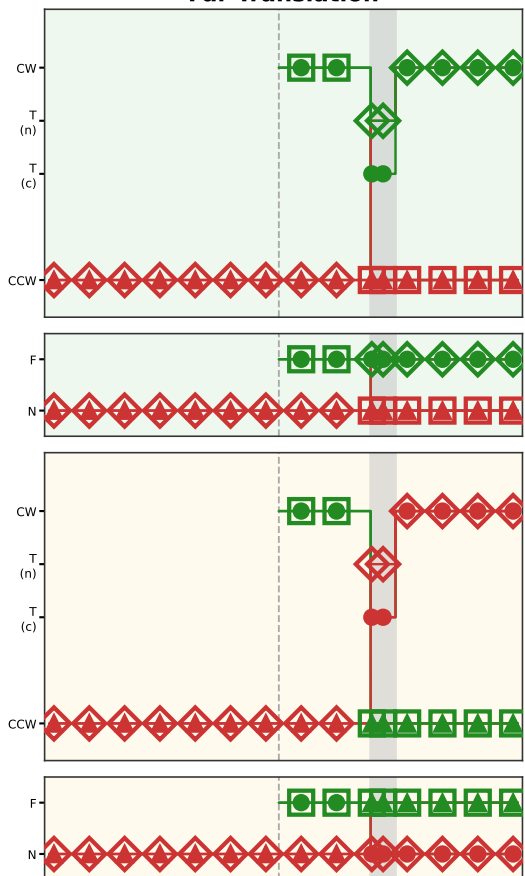
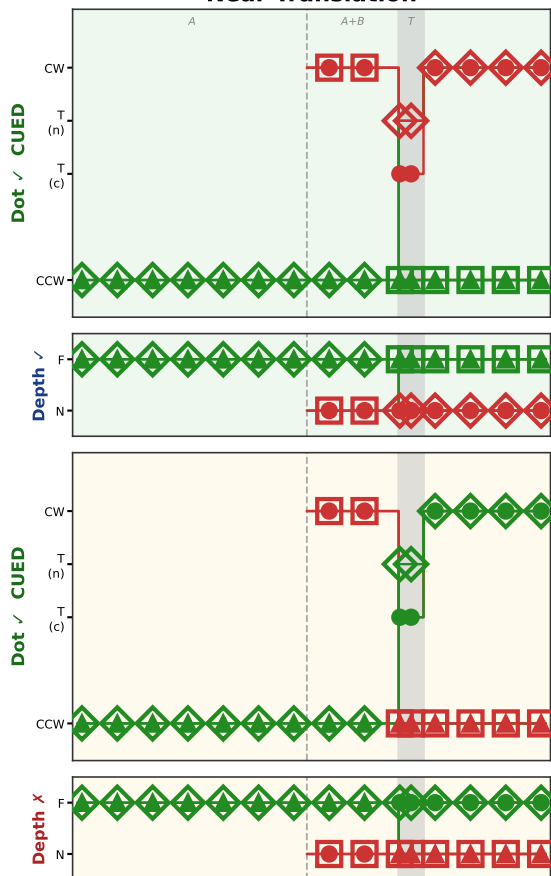
Factor 2 — Depth Cueing (✓/✗): does translation occur at the depth where the cued field appeared?

Near plane = Red · Far plane = Green · Color changes at T_s for swapping subfields · Filled = coherent (S0●, S2▲) · Open = noise (S1□, S3◇)

Near Translation

Far Translation

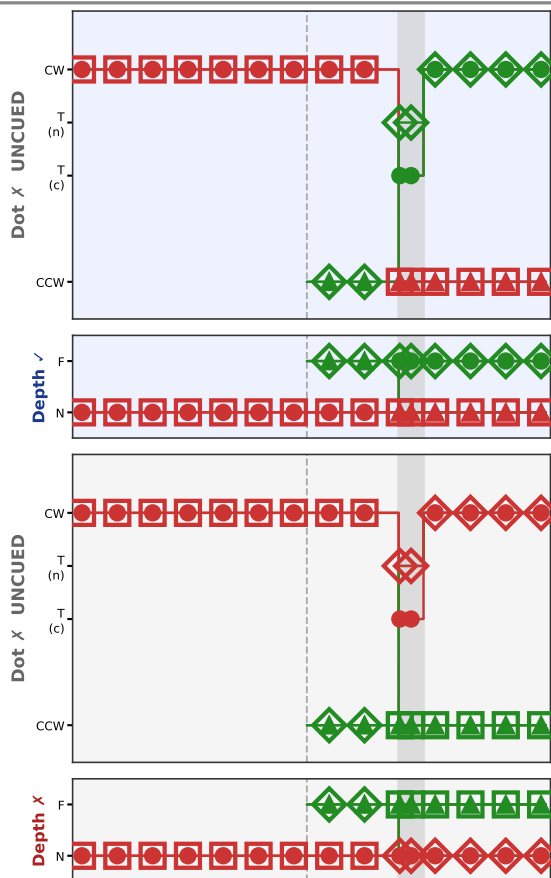
DOT
CUED



CUED + ZdNoi

CUED + ZdCoh

DOT
UNCUED



UNCUEd + ZdCoh

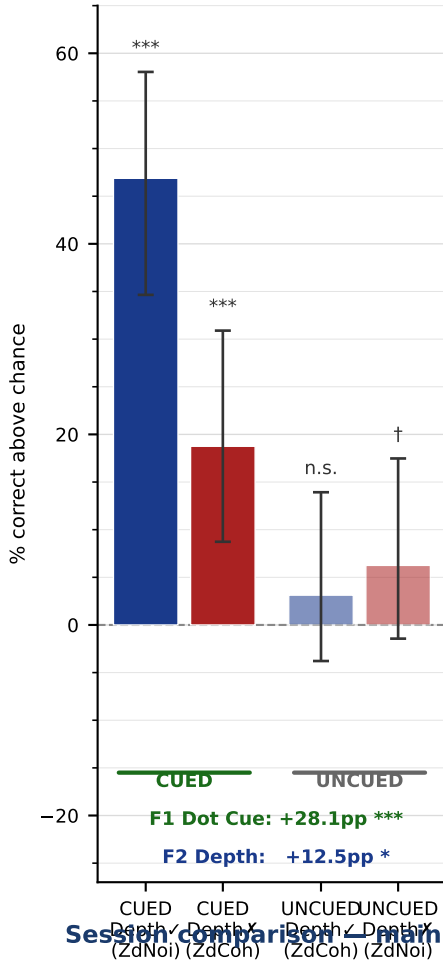
UNCUEd + ZdNoi

● S0● coh (Near=Red) □ S1□ noise (Near=Red) ▲ S2▲ coh (Far=Green) ◇ S3◇ noise (Far=Green) Translation window

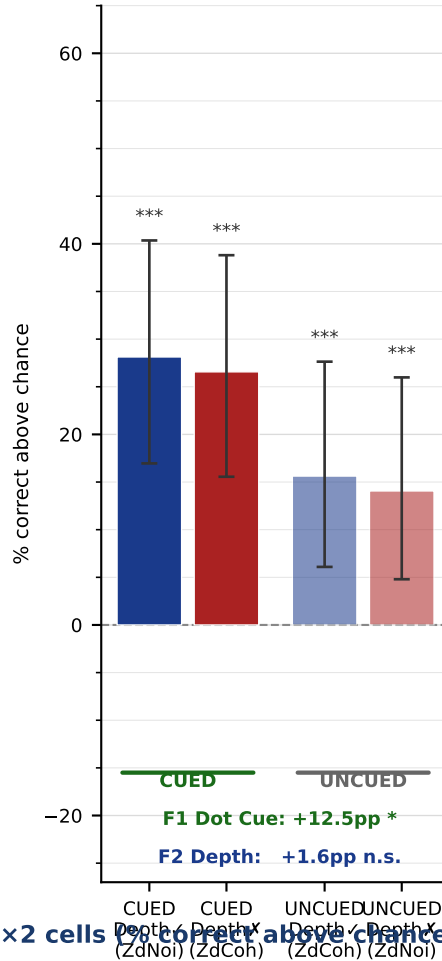
DepthColorLinked_005m — Results

Near=Red · Far=Green · depth sep = 0.05 m · 8-AFC · Bars = % correct above chance (12.5%) with 95% Wilson CI

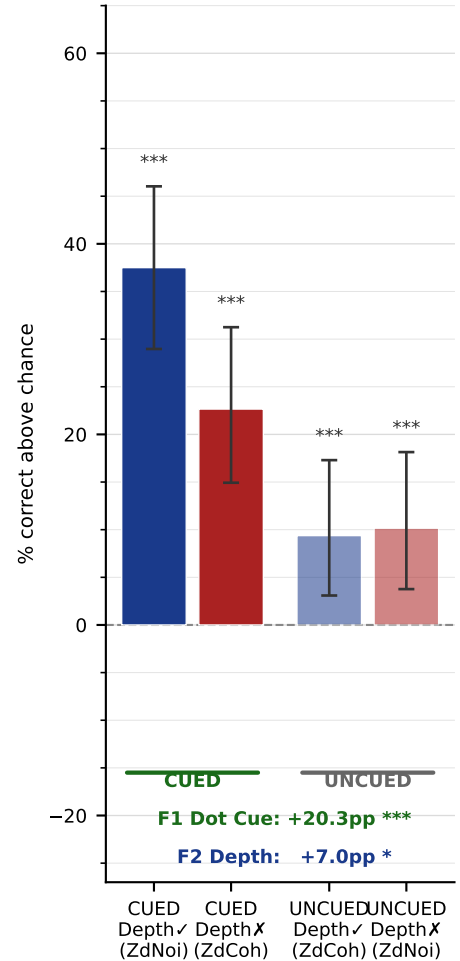
Session 1 (260404_0940)
(n=256)



Session 2 (260404_1123)
(n=256)



Combined (n=512)



Session comparison — main 2x2 cells (% correct above chance)

Condition	Session 1	Session 2	Combined	sig (comb)
CUED Depth✓ (ZdNoi)	+46.9pp (38/64)	+28.1pp (26/64)	+37.5pp (64/128)	***
CUED DepthX (ZdCoh)	+18.8pp (20/64)	+26.6pp (25/64)	+22.7pp (45/128)	***
UNCUEDE Depth✓ (ZdCoh)	+3.1pp (10/64)	+15.6pp (18/64)	+9.4pp (28/128)	***
UNCUEDE DepthX (ZdNoi)	+6.2pp (12/64)	+14.1pp (17/64)	+10.2pp (29/128)	***
F1 Dot Cueing (CUED–UNCUEDE)	+28.1pp ***	+12.5pp *	+20.3pp ***	
F2 Depth Cueing (Depth✓–DepthX)	+12.5pp *	+1.6pp n.s.	+7.0pp *	

DepthColorLinked_005m — Interpretation & Notes

Key Findings

F1 Dot Cueing: +20.3pp *** (combined, n=512). Cued trials substantially outperform uncued across both sessions. This is the core temporal-onset cueing effect and replicates robustly.

F2 Depth Cueing: +7.0pp * (combined, p=0.045). When the translation occurs at the depth plane the cued field occupied, performance is better. This replicates the ZdA vs ZdB dissociation seen in DepthSwapCtrl (all-red).

Best condition: CUED+ZdNoi (Dot✓, Depth✓) = +37.5pp *** combined. Translator keeps its depth-plane identity and color — both cueing signals align.

Worst condition: UNCUEDEither ≈ +9–10pp (near chance). Depth/color plane identity alone provides no meaningful benefit without the temporal onset cue — dots win over depth.

Between-Session Variability

Session 1 showed a clear ZdNoi >> ZdCoh dissociation (+47pp vs +19pp CUED), with UNCUEDE near chance (+3–6pp). Session 2 was flatter: ZdNoi ≈ ZdCoh (both ~+28pp CUED), and UNCUEDE came up substantially (+14–16pp).

The UNCUEDE rise in Session 2 is suspicious. Possible causes: (1) response criterion shift — more liberal/biased responding; (2) strategy change — less reliance on temporal onset cue, more on global motion grouping; (3) sampling noise at n=64/cell.

Error analysis (Session 1): reversal rate CUED=6.2% < chance (12.5%), UNCUEDE=12.5% = chance. Heading distribution perfectly uniform (32/direction). No labeling issues detected.

Response bias: 270° direction over-represented (52 vs expected 32 in S1). Likely a motor/pointing bias, not a labeling artifact. Monitor across sessions.

Design Notes

Depth separation: 0.05 m at 2 m viewing distance (~1.4° disparity at aperture edge). Chosen from DepthParam parametric sweep; Far cueing robust at all depths tested (0.03–0.15 m). Near cueing noisier — may relate to monocular positional-shift confound that scales with eccentricity.

Color-depth confound: within this experiment, depth change and color change always co-occur (linkDepthColor=1). ZdCoh = depth+color change; ZdNoi = neither. Cannot separate the two within this design alone.

Cross-experiment comparison to DepthSwapCtrl (all-red, same depth): DepthSwapCtrl ZdA CUED = +12.5pp (bino, n=64); DepthColorLinked ZdA CUED (S1) = +18.8pp. Difference hints that adding color makes ZdCoh less disruptive, but sessions differ in baseline so interpretation is premature. Need matched n.

Factor Map

F1 Dot Cueing	CUED vs UNCUEDE — does the delayed-onset dot field translate?
F2 Depth Cueing	Depth✓ vs Depth✗ — does translation occur at the DFD during T?
Depth✓	= CUED+ZdNoi OR UNCUEDE+ZdCoh (translator at DFD during window)
Depth✗	= CUED+ZdCoh OR UNCUEDE+ZdNoi (translator at opp(DFD))
F3 TransDepth	Near vs Far — depth of translating field during window (post-hoc)
F4 TransColor	R vs G — color of translating field (perfectly correlated with F3)
F5 CuedColor	R vs G — color of delayed onset field (post-hoc)
F6 RotCfgr	CW vs CCW rotation of delayed field (post-hoc, expected null)

Planned Next Steps

- More sessions: n=64/cell → target n≥128/cell for interpretable secondary factors.
- Cross-experiment comparison at matched n: DepthColorLinked vs DepthSwapCtrl (all-red) → isolate color contribution to ZdCoh disruption.
- Color-only swap experiment (linkDepthColor=0, no depth change): complete the 2×2 of depth-only × color-only.
- Error analysis across sessions: track UNCUEDE accuracy and 270° bias for criterion/strategy monitoring.